An Effective Dual-Layer Video Understanding Architecture for Generating Audio Descriptions

Tzung-Pei Hong
Department of Computer Science and Information Engineering
National University of Kaohsiung
Kaohsiung, Taiwan
tphong@nuk.edu.tw

Peng-Yu Cheng
Department of Computer Science and Information Engineering
National University of Kaohsiung
Kaohsiung, Taiwan
a1105535@mail.nuk.edu.tw

Chuan-Kai Liu
Department of Computer Science and Information Engineering
National University of Kaohsiung
Kaohsiung, Taiwan
a1105522@mail.nuk.edu.tw

Yin-Jhe Jhuang
Department of Computer Science and Information Engineering
National University of Kaohsiung
Kaohsiung, Taiwan
a1105513@mail.nuk.edu.tw

2025 IEEE Gaming, Entertainment, and Media Conference (GEM) | 979-8-3315-7517-5/25/$31.00 ©2025 IEEE | DOI: 10.1109/GEM66882.2025.11155674

Abstract— **This research introduces an innovative platform for audio-description production using multimodal large language models. Centered on the proposed 5W1H dual-layer video understanding architecture, the system integrates scene detection, text-to-speech mechanism, and audio mixing to reduce production costs and enhance accessibility for individuals with visual impairments. The first layer employs the 5W1H (who, what, when, where, why, and how) method to generate relevant prompts for extracting key information, while the second layer refines these outputs into precise descriptions. Benchmark evaluations on Video-MME reveal significant improvements in temporal reasoning, temporal perception, and object recognition compared to the baseline model.**

Keywords—Large Language Model, Audio Description, Gemini 1.5, 5W1H, Prompt

I. INTRODUCTION

Audio description (AD) is a service that helps visually impaired individuals enjoy audio-visual programs. Without interfering with the program's original audio and dialogue, audio description interprets and describes visual elements in a film, such as spatial arrangements, settings, facial expressions, and movements. These descriptions are then recorded and post-produced to create an audio-description version of the soundtrack, which integrates the original film with the narration for repeated playback [1].

Human audio describers undergo specialized training to accurately interpret visual content and adapt to different contexts and needs. For certain visual elements, specific domain knowledge may also be required. To provide the best visual information, rigorous and meticulous preparation procedures are essential. Therefore, professional audio description is a highly time-intensive and labor-intensive service [1]. Besides, producing audio descriptions requires high-level expertise. Scriptwriters must repeatedly watch films, document key scenes, and draft scripts that are both fluent and descriptive. They must also synchronize their narration with the film's pacing to ensure high coherence. However, the industry faces a shortage of skilled professionals due to high production costs. The National Communications Commission in Taiwan mentions that 45.1% of individuals aged 16 and above have watched online streaming video (OTT TV). Among the top ten services with the highest subscription numbers, only Netflix and Public Television Service (PTS+) offer narrated images [2]. In recent years, emerging technologies such as generative AI have gained significant momentum, and various industries are exploring the potential applications of integrating this technology. If a generative AI-based system for narrated image production can be developed, it would significantly reduce labor and financial costs, as well as decrease production time.

This study aims to investigate the feasibility of multimodal large language models (MLLMs) for generating described videos. Specifically, we first segment the video frames using Pyscenedetect [3] for MLLMs analysis and enhance scene understanding using the Haystack [4] framework along with our proposed 5W1H dual-layer video understanding architecture. Then, the MLLMs generate the descriptive script, which can be manually adjusted through the system interface. Finally, FFmpeg [5] is used for audio mixing and editing to produce the final video output. The findings of this research will provide insights into the effectiveness of using multimodal models for generating described video scripts, offering potential applications for the media industry and serving as a reference for AI developers.

II. RELATED WORK

Multimodal large language models (MLLMs) are artificial intelligence models capable of processing and understanding multiple data modalities, including text, images, audio, and video. Compared to traditional large language models (LLMs) that rely solely on text, MLLMs integrate information from different modalities, enhancing cross-modal reasoning and content generation capabilities. Prominent examples of MLLMs include GPT-4V [6], Gemini [7], and VideoLLaMA [8], which are widely applied in tasks such as image captioning, multimodal question answering (VideoQA), and video understanding.

Current video understanding approaches primarily adopt frame sampling techniques. For instance, MM-Vid [9] and LLM-AD [10] process video content by generating frame-by-frame captions using GPT-4V [6], incorporating speech recognition, and ultimately condensing the information into a summary. Experimental results indicate that, compared to pre-segmenting videos before processing, using MLLMs that can directly take raw video input yields superior comprehension performance.

In the past, due to the lack of vision-language large models (VLLMs) capable of handling video input, most video understanding methods relied on static frame extraction for content analysis. However, recent advancements in VLLMs have led to models that support native video input, such as Gemini. According to the Video-MME benchmark [11], Gemini 1.5 Pro [7], which supports raw video input, outperforms models of similar scale that only process static images, such as GPT-4o [12], in video understanding tasks. These findings suggest that MLLMs with native video input capabilities offer more comprehensive and accurate video content comprehension.

III. THE PROPOSED METHOD

This research aims to establish an automated audio description generation system based on multimodal large language models. The study will follow the existing workflow, which is divided into four parts: preparation, script creation, recording, and film production. By introducing technologies such as multimodal large language models, scene detection, and the proposed 5W1H dual-layer video understanding architecture (5W1H-DVUA), the production process will be optimized, as shown in Fig. 1. This research will focus on optimizing the most time-consuming stage of the audio description process, script creation, with an emphasis on the preparation and script creation phases. Improvements in recording and film production are expected to be addressed in future studies.

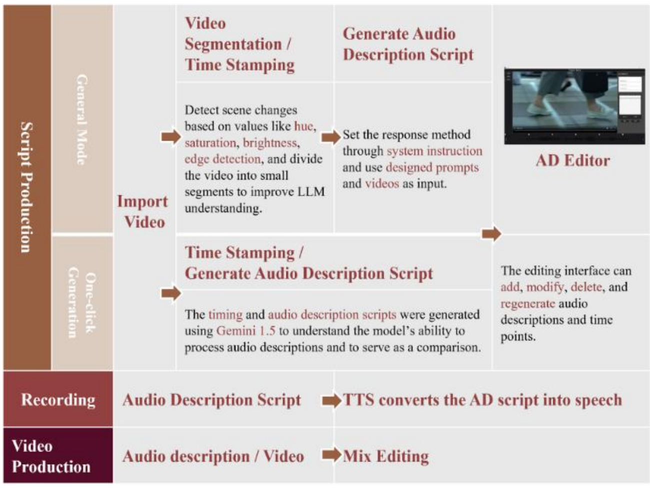

Fig. 1. Technical Flowchart

A. *Preliminary Preparation: Segmenting Videos and Using Large Language Models to Analyze Clips in Advance*

The preliminary preparation phase is crucial for creating audio descriptions. In the current production process, scriptwriters must repeatedly watch the video to understand, integrate, and absorb information about the people, events, times, places, objects, and specialized knowledge presented in the film. Existing MLLM systems for video still suffer from information loss. Through iterative reasoning, this study addresses the lack of information caused by single-viewing limitations. In the preparation stage, prior inference is used to extract key information from clips, enhancing the understanding capacity of large language models and optimizing system performance.

1) Video analysis*:* Before submitting the video to MLLMs for content interpretation, scene transitions are detected using the PySceneDetect, based on the Video-MME benchmark for video understanding in MLLMs proposed by Fu, et al. [11]. These standards indicate that LLMs perform better with short-to-medium-length video. The process involves converting each video frame into the HSL color space, assigning weights to features such as hue, saturation, lightness, and edge detection results, and calculating a score for each frame. When a frame's score exceeds a set threshold, the system identifies it as a scene transition point. However, a single fixed threshold might not yield ideal results since different videos may have varying visual styles. To address this, we introduce an adaptive scene detection mechanism that uses a rolling average to adjust the threshold dynamically. It enables the system to adapt to different scene transition patterns, ensuring sufficient accuracy across diverse scenarios and improving the content comprehension of LLMs for various video types.

2) Enhancing video understanding: Currently, most methods for improving the performance of MLLMs rely on retraining to adjust model parameters for enhanced video comprehension. However, this approach requires extensive data and computational resources, and most commercial models do not support retraining with video data. Therefore, this study introduces the 5W1H dual-layer video understanding architecture, which enhances performance without requiring additional training, reducing training costs and resource consumption. This method is also flexible, allowing it to adapt to various video content and requirements, making it more efficient and scalable in practical applications.

The first layer is the 5W1H (6W) analysis for video understanding. This layer aims to help MLLMs build a structured understanding of video content. By applying the 5W1H method, the video is analyzed from multiple perspectives, converting key information from the video frames into structured natural language descriptions. This layer generates a JSON file containing the 6W description based on the video content, with each field corresponding to characters, events, time, location, reasons, and processes within the video. This enables the model to transform video content into text.

The second layer generates video summaries and Q&A-based audio descriptions. Based on the video summary generated in the first layer, this layer further optimizes and produces professional audio description scripts. The model integrates the summary with video information, generating audio descriptions that meet industry standards by detailing scenarios, character interactions, and key elements in the video. It also ensures the structured output is both standardized and scalable. This layer introduces a multi-round Q&A mechanism to perform in-depth reasoning and supplementation on details, emotions, and causal relationships within the video, thereby enhancing the accuracy and narrative quality of the audio description.

B. *Script Production: Utilizing Video Analysis Technology and Multimodal Large Language Models to Draft Audio Description Scripts*

In the current audio description production process, scriptwriters must repeatedly watch videos to extract critical visual information and faithfully reproduce it in spoken language while staying true to the original work. This study leverages video analysis technology and script data to detect scenes in the video, ensuring that the audio description does not interrupt during transitions. Using images and text as inputs, multimodal generative models are applied to analyze scenes and draft and refine audio description scripts. The flowchart for script production is shown in Fig. 2, with each module explained below.

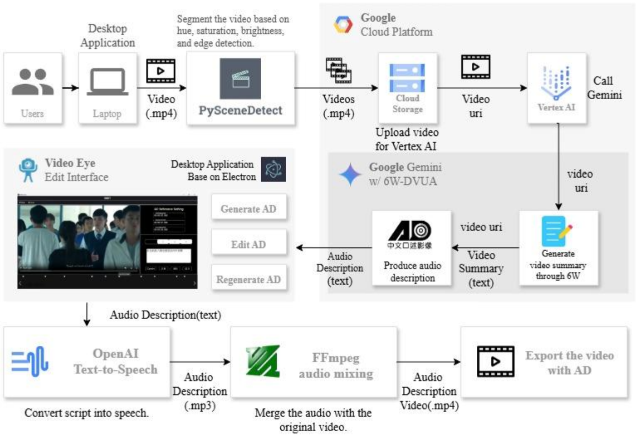

Fig. 2. System Workflow Diagram

1) Drafting audio description scripts with multimodal large language models: This study utilizes models such as Gemini Pro and Gemini Flash, along with prompts designed based on 26 principles [13]. Keyframes, script information, and prompts are inputs, and the temporal window size is adjusted to identify the optimal input and model combinations.

2) Application of a workflow management system: The study utilizes Haystack [4] as part of the workflow management system to enhance the efficiency, accuracy, and flexibility of system operations. Haystack offers a modular structure and a pipeline architecture, enabling seamless integration across processing stages and dynamic adjustments to system configurations. Haystack's pipeline architecture also allows the interconnection of various processing components, making the overall workflow more adaptable.

3) Manual modification using the system: The system is designed with the Electron framework [14], allowing users to upload video files and generate audio description scripts through the interface. If the generated transcript is unsatisfactory, users can manually adjust it, making the system more practical and user-friendly.

4) Audio mixing and editing to generate videos with audio descriptions: After obtaining the audio description scripts, Cyanreg proposes using FFmpeg [5] to mix the completed audio description file with the original video and adjust the audio levels to ensure a balanced output between the two tracks. The process involves the following steps. First, both the audio description file and the input video's audio track are extracted. The input video's audio track is saved separately to avoid modifying the original audio directly. Then, the extracted audio tracks are merged, and the volume levels of the audio description track are adjusted as needed. Finally, the merged audio replaces the original audio track in the video.

IV. Experiments

The study employed the Gemini-1.5-Flash-001 model [7] to test the performance of 5W1H-DVUA on the Video-MME benchmark [11], validating the effectiveness of this framework. To ensure a focused and controlled evaluation, we use only the short-video subset of Video-MME. The experimental results for Gemini 1.5 Flash are shown in Table I, and for Gemini 1.5 Pro are shown in Table II. They indicate that applying this framework could improve question-answering performance by 20% in temporal reasoning and increase time perception by 7%, while object recognition performance saw a 2% improvement.

TABLE I. Performance Comparison Between Baseline Gemini-1.5-Flash and 5W1H-DVUA-Enhanced Gemini-1.5-Flash on Video-MME Benchmark

Tasks	Gemini 1.5 flash	Gemini 1.5 flash w/ 5W1H-DVEUA
Temporal Perception	83.30%	88.90% (+6.72%)
Spatial Perception	80.00%	76.70% (-4.13%)
Attribute Perception	90.20%	88.50% (-1.88%)
Action Recognition	83.20%	79.20% (-4.18%)
Object Recognition	81.50%	83.30% (+2.21%)
OCR Problems	91.20%	91.20% (0.00%)
Counting Problem	50.40%	57.60% (+14.29%)
Temporal Reasoning	76.90%	92.30% (+20.03%)
Spatial Reasoning	92.60%	88.90% (-4.00%)
Action Reasoning	76.60%	80.90% (+5.61%)
Object Reasoning	80.00%	81.00% (+1.25%)
Information Synopsis	97.60%	92.70% (-5.02%)
Overall	80.60%	81.10% (+0.62%)

TABLE II. Performance Comparison Between Baseline Gemini-1.5-Pro and 5W1H-DVUA-Enhanced Gemini-1.5-Pro on Video-MME Benchmark

Tasks	Gemini 1.5 Pro	Gemini 1.5 Pro w/ 5W1H-DVEUA
Temporal Perception	88.90%	88.90% (+0.00%)
Spatial Perception	80.00%	80.00% (+0.00%)
Attribute Perception	93.30%	93.30% (+0.00%)
Action Recognition	83.20%	81.00% (-2.64%)
Object Recognition	84.50%	85.00% (+0.59%)
OCR Problems	94.70%	93.00% (-1.80%)
Counting Problem	56.80%	59.20% (+4.23%)
Temporal Reasoning	76.90%	76.90% (+0.00%)
Spatial Reasoning	96.30%	92.60%(-3.84%)
Action Reasoning	89.40%	89.10% (-0.34%)
Object Reasoning	82.50%	82.30% (-0.24%)
Information Synopsis	95.10%	97.60% (+2.63%)
Overall	83.50%	83.60% (+0.12%)

V. Conclusion

This study has successfully designed and implemented an audio description generation system by integrating multimodal large language models. After incorporating editing functionalities, the system demonstrates significant cost, efficiency, and quality advantages. The following are some key benefits of the system.

1) Cost savings: Traditional audio description production for each film costs between NT$200,000 and NT$300,000,

whereas this system requires less than NT$100 in token fees to generate audio descriptions.

2) High efficiency: While traditional methods for producing a one-hour film require approximately 2 to 3 months for audio description, this system can complete the process in under one hour.

In the future, the system may be expanded to handle videos in multiple languages to enhance its applicability. Personalized preferences can also be introduced, allowing users to adjust the content of the descriptions according to their needs, further increasing the system's flexibility.

ACKNOWLEDGMENT

This work was supported by the National Science and Technology Council (NSTC) in the College Student Research Program under the contract MOST 113-2813-C-390-008-E.

REFERENCES

[1] "Understanding audio description services for film and television," *Bureau of Audio-visual and Music Industry Development, Ministry of Culture,* 2021.

[2] "2020 convergent development survey," *National Communications Commission,* 2020.

[3] B. Castellano, "PySceneDetect," *[Online]. Available: https://www.scenedetect.com/.*

[4] "Haystack by deepset," *[Online]. Available: https://haystack.deepset.ai/.*

[5] "FFmpeg," *[Online]. Available: https://ffmpeg.org/.*

[6] OpenAI, *GPT-4 Technical Report*, 2023.

[7] Gemini Team, "Gemini 1.5: Unlocking multimodal understanding across millions of tokens of context," Google, 2024.

[8] H. Zhang, X. Li, and L. Bing, "Video-LLaMA: an instruction-tuned audio-visual language model for video understanding," *The 2023 Conference on Empirical Methods in Natural Language Processing,* 2023.

[9] K. Lin, F. Ahmed, L. Li, C. C. Lin, E. Azarnasab, Z. Yang, J. Wang, L. Liang, Z. Liu, Y. Lu, C. Liu, and L. Wang, "MM-VID: advancing video understanding with GPT-4V(ision)," *ArXiv,* abs/2310.19773, 2023.

[10] P. Chu, J. Wang, and A. Abrantes, "LLM-AD: large language model based audio description system," *ArXiv,* abs/2405.00983, 2024.

[11] F. Chaoyou, Y. Dai, L. Yondong, L. Lei, R. Shuhuai, Z. Renrui, W. Zihan, Z. Chenyu, S. Yunhang, Z. Mengdan, P. Chen, Y. Li, S. Lin, S. Zhao, K. Li, T. Xu, X. Zheng, E. Chen, R. Ji, and X. Sun, "Video-MME: the first-ever comprehensive evaluation benchmark of multimodal LLMs in video analysis," *ArXiv,* abs/2405.21075, 2024.

[12] Y. Wu, X. Hu, Z. Fu, S. Zhou, and J. Li, "GPT-4o: visual perception performance of multimodal large language models in piglet activity understanding," *Arxiv,* abs/2406.09781, 2024.

[13] S. M. Bsharat, A. Myrzakhan, and Z. Shen, "Principled instructions are all you need for questioning LLaMA-1/2, GPT-3.5/4," *ArXiv abs/2312.16171,* 2023.

[14] "Electron," *[Online]. Available: https://www.electronjs.org/.*